Project Planning

Wandr AI – AI Travel Itinerary Planner

1. Project Milestones

Phase	Milestone	Key Deliverables	Status
Phase 1 – Setup	Infrastructure provisioned	Supabase project, Groq key, Vercel + Render services	Completed
Phase 2 – Backend	Core API functional	FastAPI routes, JWT auth, ai_service.py, itinerary_service.py	Completed
Phase 3 – Frontend	UI wired to backend	All pages, AuthContext, api.ts client	Completed
Phase 4 – Security	CORS + RLS hardened	Dynamic regex CORS, Supabase RLS migration, 34-test suite	Completed
Phase 5 – Deployment	Production live	Vercel + Render deployed, health check green	Completed
Phase 6 – Docs	Documentation complete	README, SETUP.md, this documentation folder structure	In Progress

2. Sprint Breakdown (2-Week Sprints)

Sprint	Goal	Stories
Sprint 1	Foundation	Project structure, venv, FastAPI scaffold, Supabase schema migration, Vite + React scaffold
Sprint 2	Auth & CRUD	Supabase Auth integration, JWT middleware, trips CRUD endpoints, AuthContext in frontend
Sprint 3	AI Generation	ai_service.py, Groq prompt engineering, JSON parsing, itinerary display in frontend
Sprint 4	UX & Budget	Budget service, Recharts panel, EditTripPage, day regeneration endpoint

Sprint 5	Security & CI	CORS regex, RLS policies, pytest suite (34 tests), deployment pipeline
Sprint 6	Polish & Docs	Landing page, ProfilePage, README, documentation folder structure

3. Risk Register

Risk	Probability	Impact	Mitigation
Groq API deprecates model	Medium	High	GROQ_MODEL env var; model is configurable without code change
Render cold start delays user	High	Medium	Health-check endpoint warms service; upgrade to paid tier
Supabase email rate limit (free tier)	High	Medium	Disable email confirmation in dev; configure SMTP for prod
Groq rate limit on free tier	Medium	Medium	Tenacity retry; AI_RATE_LIMIT_PER_HOUR =10 per user
JWT secret rotation	Low	High	Rotate via Supabase dashboard + update Render env var
Vercel preview URL changes	High	Medium	CORS regex matches all preview URLs for the project slug